ModelIt!

Student Activity Pack

Can CRISPR Cure Genetic Diseases?

Modeling Gene Editing Precision, Off-Target Effects, and Ethical Boundaries

Grade 11L3 | NGSS HS-LS3-1, HS-LS3-2 | Cause and Effect

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you already know about how CRISPR gene editing works, and where does the technology come from?**

2. **If scientists could fix a gene that causes a fatal disease, what reasons might there be to proceed with caution?**

3. **Draw a diagram showing how you think a gene editing tool finds and changes a specific location in DNA.**

4. **What is the difference between editing genes in a person's body cells versus editing genes in an embryo?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Guide RNA Specificity | Cas9 Cutting Efficiency | Off-Target Edit Rate | Repair Pathway Selection | Delivery Efficiency | Therapeutic Benefit | Unintended Mutations

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: High-Precision Sickle Cell Edit

Maximum guide RNA specificity, optimized Cas9, single HBB target

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Low-Specificity Broad Edit

Moderate guide RNA specificity targeting a gene with many similar sequences

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Germline vs. Somatic Comparison

Same editing parameters applied to somatic versus germline cells

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Epigenetic Disruption — Changes to gene regulation patterns caused by CRISPR editing near regulatory regions — even successful target edits can alter how neighboring genes are expressed, potentially causing unexpected phenotypic effects

Immune Response — The patient's immune system reaction to Cas9 protein and viral delivery vectors — many humans have pre-existing antibodies to common Cas9 variants because they evolved from bacteria that infect humans

Mosaicism Rate — The percentage of cells that receive different editing outcomes — some cells may be correctly edited, others incorrectly edited, and others unedited, creating a genetic mosaic that complicates therapeutic efficacy

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Could we rewrite the code of life to eliminate genetic disease — and should we? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate the tension between editing precision and the sheer size of the human genome?**

2. **Why is the cell's choice between repair pathways such a critical variable, and how does it limit CRISPR's therapeutic potential?**

3. **What does your model reveal about the difference in risk profiles between somatic and germline editing?**

4. **How would your model change if a new Cas9 variant were developed with zero off-target activity?**

5. **What are the limitations of modeling gene editing when individual patient genomes vary by millions of base pairs?**

8 STEM Challenge: Design a CRISPR Clinical Trial Protocol

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YOUR ENGINEERING MISSION

Design a clinical trial protocol for using CRISPR to treat a specific genetic disease, addressing editing precision, safety monitoring, delivery strategy, and ethical safeguards.

THE SCENARIO

A biotech company has asked your team to design the Phase 1 clinical trial protocol for a new CRISPR therapy targeting Huntington's disease — a fatal neurological condition caused by a single gene mutation. You must balance the urgency of treating a terminal disease against the safety risks of off-target editing in brain tissue, design a delivery strategy that reaches neurons, and establish monitoring protocols to detect unintended mutations before they cause harm.

Engineering Design Thinking

1. WHAT GUIDE RNA DESIGN FEATURES MAXIMIZE SPECIFICITY FOR THE HUNTINGTON'S GENE WHILE MINIMIZING OFF-TARGET POTENTIAL?

2. WHAT DELIVERY METHOD WOULD YOU CHOOSE FOR BRAIN TISSUE AND WHAT ARE THE TRADE-OFFS VERSUS OTHER APPROACHES?

3. HOW MANY PATIENTS SHOULD BE IN YOUR PHASE 1 TRIAL AND WHAT ARE YOUR PRIMARY SAFETY ENDPOINTS?

4. WHAT ETHICAL REVIEW PROCESS WOULD YOU REQUIRE BEFORE THE FIRST PATIENT RECEIVES CRISPR TREATMENT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Genetic Counselors help patients and families understand genetic test results and the implications of gene therapy options, earning $85,000-$120,000/year. Gene Therapy Scientists design and test CRISPR-based treatments at biotech companies like Editas Medicine, CRISPR Therapeutics, and Intellia, earning $95,000-$180,000/year. Bioethicists who specialize in emerging genetic technologies advise hospitals, companies, and governments on ethical frameworks, earning $70,000-$140,000/year.